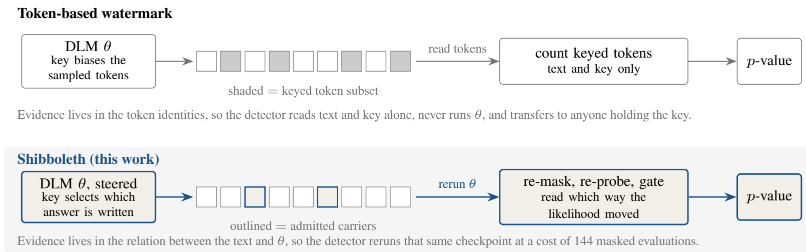

Figure 1: **Two places to keep a watermark.** A token-based watermark biases which tokens are sampled and recovers that bias by reading the text against the key. Shibboleth instead writes the answer the key requests to a keyed probe of the context, which makes the evidence a property of the text together with one checkpoint and obliges the detector to rerun it.

Shibboleth: Probe-and-Replay Model-Response Watermarking for Diffusion Language Models

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

We propose Shibboleth, a model-response watermark for masked diffusion language models (DLMs). A masked DLM can be re-evaluated on its own output under any remasking of the context, and because a practical DLM shifts the likelihood of an emitted token by an amount its own weights determine, a watermarking channel opens in the relation between a text and the model rather than in the tokens. Shibboleth evaluates a keyed family of context probes against each masked block, admits as carriers only the positions whose block-masked conditional retains mass on both response signs, and resamples at those carriers from the live conditional until the requested sign is committed, leaving the model's probabilities and the decoding schedule untouched. Because the admitting gate is invariant to swapping two probes, it is blind to the sign the key will request, so carriers may be discovered adaptively without disturbing the null. Detection re-masks the finished text, rebuilds the probes and carriers, and counts keyed sign matches, whose null distribution is exactly binomial, so p-values are exact at finite sample size under carriers the method itself chose. It reaches 0.80 TPR at 1% FPR on LLaDA-8B-Instruct — below the strongest token-based detector there, at the only perplexity ratio reading below its own unwatermarked baseline — and 0.90 on Dream-7B-Instruct, where every token-local baseline weakens under transfer and the model-response channel is the strongest detector measured. Verification costs one rerun of the checkpoint, and evidence written against a prefix degrades under dispersed editing, leaving the method suited to a verifier that holds the checkpoint rather than to open screening.

1 Introduction

Language models now write text that readers cannot reliably tell from human writing, and provenance becomes a security problem once disinformation, phishing and academic fraud stop needing an author [28]. A post-hoc detector is fitted to one generation process and weakens as deployment moves to models that decode by another mechanism, masked diffusion models among them; watermarking instead plants a keyed signal during generation and lets the key holder test a document later [21, 32]. A watermark is worth only the guarantee behind its verdict — a false positive accuses a human author — so a deployment needs a false-positive rate it can state in advance rather than fit to a corpus, and, since adversaries paraphrase and learn a key from detector queries [19, 23], one that survives finite samples and adaptively chosen trials [10, 43].

Existing watermarks place their secret in the tokens, whether by biasing keyed vocabulary subsets [21], by coupling each draw to pseudorandomness [24], or, in recent diffusion language model (DLM) methods, by exploiting unresolved context and denoising order [17]. Detection then reduces to inspecting the text together with the key, and the generator is never consulted. Masked DLMs, by contrast, permit a different arrangement, since a finished text can be fed back to the model with selected context hidden and the likelihood of an emitted token then moves by an amount the checkpoint alone determines. Hiding a keyed subset of the context is in this sense a *context probe*, the direction in which the likelihood moves is the model's response to it, and provenance can be argued from the agreement between the two. Such a probe is worth applying only where the model has a genuine choice of answer, that is where several plausible tokens coexist at a masked position and react with opposite signs to two patterns.

Moving the secret out of the tokens and into that relation, however, costs two things a token-local detector obtains for nothing, since the verifier must apply exactly the probe the embedder applied, and the choice of carriers must be settled before the accepted answer is known; a gate with sight of the rewarded sign would otherwise inflate the false-positive rate above the level the detector reports. Section 3.2 states both as requirements, and existing DLM watermarks avoid them by keeping detection token-local, whether by removing prefix dependence [9], by averaging a red–green bias over unresolved context [14], or by steering which mask is resolved next [17], none of which asks the model anything about the text it produced.

Shibboleth instead meets each condition with one mechanism (Figure 1). It settles replayability by re-masking the current block together with its suffix before scoring the probe family, and the state it measures therefore follows from the final text alone. A *balanced gate* then admits as carriers only the unresolved positions whose block-masked conditional retains mass on both response signs, scoring them in a way that is invariant to swapping the two patterns and therefore blind to the sign the key will ask for. Writing the answer is the easier half, since Shibboleth draws at most R proposals from the unchanged live row at an admitted position and commits the first one carrying that sign, which amounts to rejection sampling and does change the output distribution, although every committed token remains one the model proposed. Verification afterwards re-applies the same probe, reapplies the gate, and returns an exact binomial tail. Unlike a prefix-based watermark, which can require strict left-to-right decoding, and unlike an order-guided one, which changes which mask is resolved next, Shibboleth acts only after the reference schedule has produced a row, and therefore leaves the block partition and the confidence order untouched.

- We show across 36 settings that this checkpoint's conditionals are too sharp for answers to be inserted after the fact, and the answer must therefore be written while the conditional is still live (Section 3.2).
- We propose Shibboleth, in which block-and-suffix re-masking makes the probe replayable from the finished text and orientation-blind gating keeps the accepted answer independent of how the carriers were discovered, which together keep the detector's p-values exact and free of calibration (Sections 3.3 to 3.5).
- We evaluate Shibboleth on LLaDA-8B-Instruct and Dream-7B-Instruct and report both sides of the trade: the strongest setting reaches 0.80 and 0.90 TPR at 1% FPR — on LLaDA at the best perplexity ratio of any method compared, on Dream above every token-local baseline under shared settings — whereas verification spends 144 masked evaluations and 10% re-denoising collapses TPR at $R = 1$ and halves it at the strongest setting (Section 4).

2 Related Work

Diffusion language models. Discrete and masked diffusion objectives carry diffusion [16] over to text [3, 29, 31, 36, 37] and now support instruction-following systems such as LLaDA and Dream

[25, 30, 41, 44]. Since commit order affects quality and throughput [2, 6, 20, 26, 38, 39], accelerators treat the schedule as a tunable degree of freedom; Shibboleth treats it as an invariant and uses masked-token reconstruction only to define a probe it can later repeat.

Watermarking autoregressive text. Autoregressive watermarks bias keyed token subsets [21] or couple sampling to keyed randomness [1, 13, 24], with later work characterising their reliability and undetectability [10, 22, 43]. Detectors of this family are inexpensive precisely because a key–token hash needs no generator, yet that hash is defined over a resolved prefix and does not transfer unchanged to a context in which most token positions are still masked.

Watermarking diffusion language models. DLM-specific methods restore a well-defined keyed partition by other means, whether by removing prefix dependence [9], by averaging a red–green bias over unresolved context [14], by keyed sampling [4], or by exploiting the decoding order itself [17, 33, 40]. We compare against dgMARK, the strongest order-steering baseline with a zero-inference detector. Shibboleth differs from all of them in where the evidence resides, since dgMARK spends the DLM's freedom over commit order and reads the outcome off the text, whereas Shibboleth spends nothing from the schedule and instead has the verifier rerun the exact checkpoint, buying a statistic bound to model behaviour at the price of a detector that no longer runs on text alone.

Detector statistics. Detector design has moved past the asymptotic z-test toward exact tails [12], permutation p-values for alignment statistics [24], and union-bound corrections for windowed maxima [22]; when edits leave evidence in only a fraction of the text, sum-based pooling is provably dominated by statistics that concentrate on the surviving fraction [27]. Shibboleth's binomial tail is an exact test, and Section 4.2 measures the sparse-signal alternatives against it.

Detection that consults the model. Every DLM watermark above detects from the text and key alone, but detectors that pay more for their evidence are not new: edit-distance detection runs quadratic alignment under thousands of permutation resamples [24], model access makes the Neyman–Pearson score strictly dominate the key-only Gumbel statistic [12], likelihood-ratio detection reads both the watermarked and base conditionals [18], GaussMark keys a Gaussian weight perturbation and verifies with one forward–backward pass [7], and model-theft watermarks assume the verifier can query the suspect system [42]. Unlike GaussMark, which ships a perturbed checkpoint, Shibboleth leaves the released weights as they are and puts the key in the questions asked of them, so one deployed model serves every key.

3 Shibboleth: Watermarking by Model Response

3.1 Decoding and Notation

Masked diffusion language models differ from autoregressive models in that they predict a missing token from an arbitrary subset of revealed tokens, and the same sequence can consequently be produced under many decoding orders. Let x be a prompt and $y = (y_1, \ldots, y_N)$ the continuation to be generated, and for a revealed set $I \subset \{1, \ldots, N\}$ with a target position $i \notin I$, the model supplies a predictor $p_\theta(y_i \mid y_I, x)$ while the remaining positions carry the mask symbol, so that one sequence evaluation returns that distribution at every unresolved position at once.

Practical decoders nevertheless leave much of that freedom unused, and we study LLaDA's semi-autoregressive schedule, in which the continuation is partitioned into blocks that finish from left to right while diffusion resolves the masks inside the active block [2, 30], and the reference confidence rule decides which positions commit at each step. Everything below is therefore stated for one block B_b with the prefix F_b that precedes it.

3.2 Problem Setup

Our goal is a watermark whose detector consults the model rather than the text alone, telling a continuation generated under a secret key from one generated without it at a false-positive rate the verifier states in advance, while disturbing the decoding schedule and the output quality as little as the channel allows.

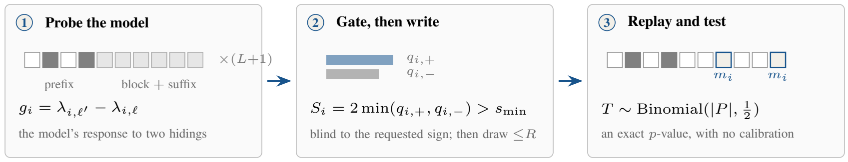

Figure 2: **How Shibboleth writes a mark and reads it back.** The same three steps run at generation and at verification, which is what makes the evidence replayable, and the gate in the middle step is settled before the key names the sign it accepts, which is what keeps its p-values exact.

Embedder and verifier. The embedder receives the model, the prompt, a secret key K, and the target bits, whereas the verifier later receives only the completed sequence, the same model, K, and the public parameters, without logits, random seeds, or commit order. Each document then draws its own key from K and a fresh nonce through a keyed hash, $K_d = \mathrm{HMAC}(K, \mathrm{nonce}_d)$, which to anyone without K behaves as a random function of the nonce [5]. That document key generates both the probe family and the accepted answers, and the false-positive guarantee covers any text produced independently of it.

Threat model. Both the model and the algorithm may be public, since secrecy rests entirely in K. Concretely, the detector does not accuse innocent text, meaning that for every text written independently of the document key, machine text from another sampler included, a detector run at nominal level α fires with probability at most α, and the bound holds at finite sample size and under carriers the method itself selects. We claim nothing in the complementary direction, and an adversary willing to edit the output can therefore suppress detection, as Section 4.2 measures. We likewise claim nothing after key disclosure, under the detector-query access that recovers a key's behaviour [19], or when verification runs a numerically different checkpoint.

Post-hoc substitution. A cheaper embedder would swap tokens in the finished text; Appendix B measures how little room such a swap has — across 36 settings the median selected position admits exactly one token at a 3-nat likelihood budget, and the best setting reaches 0.10 TPR at 1% FPR inside a $1.35\times$ perplexity budget — so the embedder must write while the conditional is still live.

We accordingly formalise the problem with three requirements.

1. **Replayability.** The verifier must be able to put exactly the question the embedder put, from the finished text and the key alone, with no generation trajectory, stored logits, or record of rejected samples.
2. **Blind selection.** The set of positions that carry a bit may depend arbitrarily on the text and the model, but that choice has to be settled before the sign the key rewards is consulted, since otherwise the reported false-positive rate understates the true one.
3. **Schedule preservation.** The block partition and the confidence order must survive unchanged, so that a production decoder keeps the latency and quality profile it was tuned for.

The first two pull against each other, since replayability wants every scored context recoverable from a text the model helped choose while blind selection wants that choice to say nothing about the sign the key rewards. Sections 3.3 to 3.5 meet each requirement with one mechanism.

3.3 Probing the Model and Gating the Carriers

Unlike dgMARK, which obtains its channel by choosing which mask is resolved next, Shibboleth leaves the decoding schedule as the reference decoder set it. For block B_b, let F_b be the prompt together with the completed prefix, and let θ denote the reference checkpoint with p_θ for its predictor. Shibboleth then constructs L equal-size near, far, and keyed pseudorandom *probe patterns* $D_{b,\ell} \subset F_b$ indexed by $\ell \in \{1, \ldots, L\}$, each the set of prefix positions it hides, and it masks $D_{b,\ell}$ together with the current block and the suffix, and one model evaluation therefore yields, for every token position $i \in B_b$ and every vocabulary item v, the log-probability

$$\lambda_{i,\ell}(v) = \log p_\theta(v \mid F_b \setminus D_{b,\ell}; B_b \text{ and suffix masked})\,. \tag{1}$$

Probing one block therefore costs $L + 1$ sequence evaluations, one per pattern plus an unprobed baseline, and the verifier later repeats the same set.

Re-masking the block and the suffix removes every token unavailable when the probe was first applied, which is what buys replayability: the verifier restores the same state from the final text alone, and a transient generation-time response becomes a document statistic. Masking only the probed prefix would fail on both sides, since token positions committed later in the same block would leak into verification-time logits while the future suffix was invisible during embedding.

For an unordered pair of pattern indices (ℓ, ℓ') and a candidate token z, define the contrast $g_i^{\ell,\ell'}(z) = \lambda_{i,\ell'}(z) - \lambda_{i,\ell}(z)$. Let p_i^{base} denote the block-masked conditional, the model distribution when B_b and the suffix are masked but no probe pattern is applied. The mass on each sign and the two-sided score built from them are

$$q_{i,\pm}^{\ell,\ell'} = \sum_z p_i^{\text{base}}(z)\mathbf{1}[\pm g_i^{\ell,\ell'}(z) > 0], \qquad S_i^{\ell,\ell'} = 2\min(q_{i,+}^{\ell,\ell'}, q_{i,-}^{\ell,\ell'}). \tag{2}$$

Shibboleth selects, independently at each token position, the pair maximizing $S_i^{\ell,\ell'}$ and stores its contrast as g_i in the bank $\mathcal{R}_b$, after which the gate admits position i whenever $S_i > s_{\min}$, subject to an optional per-block cap. Because $S_i^{\ell,\ell'}$ is unchanged when ℓ and ℓ' are swapped, the gate neither observes nor favours the orientation the key will request, which is what buys blind selection, and both the pair selection and the gating recompute from the block-masked state. Best-pair selection is moreover what makes the gate productive, raising the measured mean score from 0.28 for a fixed near/far pair to 0.45, although the score stays a symmetric proxy computed on the block-masked conditional while generation runs from an evolving live one.

Once the gate has admitted a token position, the key supplies an orientation $\epsilon_i \in \{-1, +1\}$ from a PRF domain separate from pattern generation, pair selection, carrier assignment, and payload grouping. A target $w_i \in \{-1, +1\}$ then carries the desired provenance or message bit, so that for a candidate z the shared embedding-and-verification predicate becomes

$$m_i(z) = \mathbf{1}[w_i \epsilon_i g_i(z) > 0]. \tag{3}$$

Should $g_i(z)$ vanish exactly, a second domain-separated fair bit defines $m_i(z)$, and although such ties cannot be steered, the tie rule keeps the detector's p-values exact.

3.4 Writing the Requested Answer

Shibboleth inherits the reference confidence order together with the live conditional p_i, and the token positions that the gate rejected are sampled normally. At an admitted carrier define

$$A_i = \left\{ z : w_i \epsilon_i g_i(z) > 0 \ \text{ and } \ p_i(z) \geq \kappa \max_v p_i(v) \right\}. \tag{4}$$

It draws from p_i up to R times and commits the first member of A_i, and after R misses it commits one fresh unchecked draw instead. Because the row is never recomputed, those retries cost no model evaluation.

That procedure changes the sampling distribution, and by a stated amount. Let $a_i = p_i(A_i)$ be the acceptance mass and let $q_i = \sum_z p_i(z) m_i(z)$, where $m_i(z)$ applies Eq. (3) and the tie rule to candidate z, so that q_i is the probability that an unconditional draw scores as a detector hit irrespective of the plausibility floor. The committed-token distribution is

$$\left(1 - (1 - a_i)^R\right) p_i(\cdot \mid A_i) + (1 - a_i)^R p_i, \tag{5}$$

and its detector-hit probability is

$$\Pr[m_i = 1] = 1 - (1 - a_i)^R (1 - q_i). \tag{6}$$

Whenever $\kappa = 0$ and the conditional places no mass on exact ties, $a_i = q_i$ and Eq. (6) collapses to $1 - (1 - q_i)^{R+1}$, so that a_i and q_i diverge only under a floor or a tie rule, where an unchecked fallback may score as a hit without belonging to A_i. The floor meanwhile bounds each guided proposal's model NLL to within $-\log \kappa$ of the live argmax but constrains neither the fallback nor semantic quality, which is why being model-proposed remains a support statement rather than a distortion-free claim.

3.5 Watermark Detection

Given a completed text y, the verifier re-masks each block and suffix, rebuilds the probe from the fixed prefix, reapplies the gate, and evaluates $m_i = m_i(y_i)$ at the admitted positions, so that rejected proposals and the generation trajectory are irrelevant. For provenance-only detection, which is all we evaluate here, the admitted carriers form a presence pool P with $w_i = +1$, whereas a payload implementation would keep the group for each payload bit separate, so that a balanced message cannot cancel positive against negative targets. The presence statistic and its one-sided p-value are

$$T = \sum_{i \in P} m_i, \qquad p_{\mathrm{det}} = \Pr\big[\mathrm{Binomial}(|P|, \tfrac{1}{2}) \geq T\big]\,. \tag{7}$$

Proposition 1 (Exact false-positive control)**.** *Assume the PRF domains above behave as independent random functions. Fix any text y written without the document key, and condition on the carrier set, selected pairs, contrast values, payload roles, and target bits. The orientation bits (and tie bits where needed) remain independent fair coins. Consequently the m_i are independent* $\mathrm{Bernoulli}(\frac{1}{2})$ *variables and $T \sim \mathrm{Binomial}(|P|, \frac{1}{2})$ exactly, so that p_{det} in Eq. (7) is an exact p-value and thresholding it at α holds the false-positive rate at or below α for every document length and carrier count.*

The conditioning carries the result, since carrier count, selected pair, and contrast magnitude may all depend arbitrarily on the text and the model, yet each is fixed before the independent orientation bit is consulted. Flipping ϵ_i therefore flips m_i whenever $g_i(y_i) \neq 0$, the tie rule supplies a fresh fair bit when it vanishes, and the carriers stay independent of one another throughout. In particular, neither an asymptotic approximation nor a calibration corpus enters anywhere, unlike a z-score whose reference distribution has to be estimated. The guarantee does nevertheless assume text written without the key, together with independence among the several uses of the key, which the domain-separated hash supplies.

4 Experiments

Datasets and setup. We evaluate `GSAI-ML/LLaDA-8B-Instruct` [30] and Dream-7B-Instruct [41] in fp16 on one TITAN RTX (Dream predicts position i at raw-logit index $i-1$; one shift in the wrapper restores the shared convention for generation and replay). Detectability uses C4 `realnewslike` [35] prompts truncated to 300 characters as in dgMARK. Every row decodes 512 tokens in 32-token blocks at one denoising step per token under its own regime: Shibboleth and dgMARK on the native confidence schedule with multinomial sampling at temperature 0.8 (dgMARK with top-k=3), KGW strictly left-to-right as its green list requires. Shibboleth adds $L = 8$ probe patterns, threshold $s_{\min} = 0.5$, retry budget R and an optional floor κ; we test provenance detection $(w_i = +1)$, not payload recovery.

Metrics and baselines. Detectability is TPR at analytic FPR thresholds from Eq. (7); quality is GPT-2-large [34] perplexity against an unwatermarked baseline from the same decoder, $\exp(\mathrm{mean}(\mathrm{NLL}_{\mathrm{wm}} - \mathrm{NLL}_{\mathrm{ctrl}}))$, which measures each method's distortion of its own baseline rather than absolute quality, so ratios compare across methods only indicatively. Compute counts sequence evaluations: each of the $512/32 = 16$ blocks needs $L + 1 = 9$, hence 144 for verification and $512 + 144 = 656$ for generation. Baselines are KGW [21] under left-to-right decoding and dgMARK [17] under confidence decoding.

4.1 Main Results and Analysis

Retry budget and the floor. Detection scales with the number of guided draws as Eq. (6) predicts: $R = 1$ reaches 0.43 TPR at 1% FPR, and $R = 2$, 4, 8 on 30 held-out prompts reach 0.73, 0.83, 0.87. Retries cost no model evaluation, so the budget is paid in quality — unfloored perplexity ratios climb from $1.14\times$ at $R = 2$ to $1.61\times$ at $R = 8$ — and the floor converts part of it back: $(R, \kappa) = (8, 0.1)$ obtains 0.70 TPR at $0.87\times$ and $(16, 0.05)$ obtains 0.80 at $0.74\times$, both below their unwatermarked baselines, although a ratio below one says only that GPT-2-large prefers the high-probability tokens the floor admits.

Across models. Carrying every method's LLaDA-chosen setting to Dream unchanged inverts the ranking: KGW falls to 0.57 TPR at 1% FPR at $1.84\times$ perplexity, since forced left-to-right decoding

Method	Venue	Setting	PPL ratio ↓	TPR at FPR ↑ 1%	0.1%	Sequence evaluations ↓ Generation	Detection
LLaDA-8B-Instruct							
KGW [21]	ICML'23	$\delta = 1$, 512 tok	1.22×	0.90	0.83	512	**0**
dgMARK [17]	ICML'26	$k = 1$, 512 tok	1.45×	0.83	0.80	512	**0**
dgMARK [17]	ICML'26	+3-beam, 512 tok	1.21×	**0.93**	**0.93**	∼2048	**0**
Shibboleth	ours	$R = 1$, 512 tok	0.97×	0.43	0.20	656	144
Shibboleth	ours	$R = 8, \kappa = 0.1$, 512 tok	0.87×	0.70	0.60	656	144
Shibboleth	ours	$R = 16, \kappa = 0.05$, 512 tok	**0.74**×	0.80	0.60	656	144
Dream-7B-Instruct							
KGW [21]	ICML'23	$\delta = 1$, 512 tok	1.84×	0.57	0.37	512	**0**
dgMARK [17]	ICML'26	$k = 1$, 512 tok	1.75×	0.37	0.27	512	**0**
dgMARK [17]	ICML'26	+3-beam, 512 tok	**0.71**×	0.30	0.30	∼2048	**0**
Shibboleth	ours	$R = 1$, 512 tok	1.34×	0.40	0.27	656	144
Shibboleth	ours	$R = 8, \kappa = 0.1$, 512 tok	0.79×	0.70	0.67	656	144
Shibboleth	ours	$R = 16, \kappa = 0.05$, 512 tok	1.09×	**0.90**	**0.90**	656	144

Table 1: **Watermark detectability.** Detection rate, perplexity against each method's own matched unwatermarked baseline, and compute in sequence evaluations; all rows are local reproductions at 512 tokens, each method under its own decoding regime, so PPL ratios compare within a method and only indicatively across (< 1 is not better semantics). Shibboleth rows use $n = 30$ documents ($n = 20$ for $R = 16$); Dream PPL ratios rest on the 8–13 pairs per arm that survive the decoded-length rule. δ is KGW's logit bias, k dgMARK's candidate count. Every method keeps its LLaDA-chosen setting on Dream, so cross-model rows measure transfer, not per-model optima.

Model	Method	MMLU Acc ↑	GSM8K Acc ↑	HumanEval Pass@1 ↑
LLaDA-8B-Instruct	Non-watermarked	0.595	0.640	0.335
	KGW [21]	0.535	0.740	0.293
	dgMARK [17]	0.530	0.660	??
	dgMARK +3-beam	0.530	0.660	??
	Shibboleth	0.575	0.700	0.384
Dream-7B-Instruct	Non-watermarked	0.695	0.660	??
	KGW	0.625	0.700	??
	dgMARK	0.570	0.560	??
	dgMARK +3-beam	0.645	0.560	??
	Shibboleth	0.710	0.680	??

Table 2: **Text generation quality** under multinomial decoding: MMLU [15] (200 questions), GSM8K [11] (50 problems) and HumanEval [8] (164 tasks, executed). Shibboleth uses $R = 16, \kappa = 0.05$. Each watermark decodes under its own regime — KGW left-to-right, dgMARK under confidence decoding — and is read against its own control (Appendix A), not across rows; differences under one standard error (0.035 on MMLU, 0.07 on GSM8K, 0.037 on HumanEval) are noise. Greedy-decoding cells are in Table 5.

makes Dream repeat itself (84% duplicate bigrams), and dgMARK falls to 0.37 direct and 0.30 with lookahead, whereas $(16, 0.05)$ reaches 0.90 at both 1% and 0.1% FPR (sync 0.725 against 0.503 on a control with zero false positives) and is the strongest detector on this model. A token-local channel is only as strong as the decoder behaviour it exploits; a probe the model answers about its own output travels with the model. The floor's perplexity dividend does not transfer — $(8, 0.1)$ reads 0.79× but $(16, 0.05)$ 1.09×, on 8–13 surviving pairs per arm — so the quality-optimal (R, κ) is model-dependent. No method was tuned per model.

Downstream quality. Perplexity is a proxy, so Table 2 measures accuracy on GSM8K (50 problems, 256 tokens), MMLU (200 questions) and HumanEval (164 tasks, executed) under multinomial decoding, each watermark against its own control. Shibboleth never leaves the noise band of its control on either model (GSM8K 0.700 vs. 0.640 and 0.680 vs. 0.660; MMLU 0.575 vs. 0.595 and 0.710 vs. 0.695; HumanEval 0.384 vs. 0.335 on LLaDA), so the sub-control perplexity is not bought with degenerate text. dgMARK pays for its channel on Dream — 0.840 to 0.560 on GSM8K in both variants and 0.740 to 0.570 on MMLU, well beyond noise — and 0.075 on LLaDA MMLU. KGW leaves accuracy untouched but barely embeds on short constrained answers (mean $z = +0.46$, zero TPR on LLaDA GSM8K). Short task outputs starve every detector: Shibboleth and direct dgMARK both read 0.12 TPR at 1% FPR on LLaDA GSM8K and three-beam dgMARK 0.44, so headline TPR at 512 tokens transfers to 256-token task outputs for no method.

Cost against token-local detectors. Detection trails the token-based detectors at matched length — three-beam dgMARK reaches 0.93 at both 1% and 0.1% FPR against 0.80 and 0.60 for Shibboleth, direct dgMARK 0.83 and 0.80, KGW 0.90 and 0.83 — while Shibboleth alone reads below its own unwatermarked baseline in perplexity, $0.74\times$ against $1.22\times$, $1.45\times$ and $1.21\times$, since the floor discards exactly the low-probability draws a proxy scorer punishes. KGW also needs strict left-to-right decoding, under which this checkpoint repeats itself, and its repeated bigrams must be deduplicated: undeduplicated, its control measured 37% FPR, and even deduplicated two of thirty 512-token null documents cross the 0.1% threshold — the residual calibration failure an exact conditional null rules out by construction [12]. Shibboleth alone spends 144 masked evaluations to detect at all; at these cohort sizes rows within about a tenth of TPR are ties.

4.2 Calibration and Robustness

Calibration under the shipped rule. Proposition 1 is analytical, which is why we also measure the rule as deployed. Across 20 keys and 120 non-watermarked generations apiece on LLaDA, the mean FPR is 0.0121, 0.0037 and 0.0000 at nominal 0.10, 0.05 and 0.01, and the worst individual key gives 0.0583, 0.0250 and 0.0000, and control consequently holds for every key rather than on average alone, which is what binds when one key serves many documents (Table 3, Appendix A). Every Dream control cohort in this section behaves the same way, with zero false positives at 1% throughout.

Post-editing. Edit robustness is the channel's principal failure mode (Table 4, Appendix A). At $R = 1$, same-model argmax re-denoising of a random 10% of positions takes TPR at 1% FPR from 0.35 to 0.05 on 20 documents, and deleting 10% measures the same: an early edit alters the prefix from which every later probe is rebuilt, so what collapses is synchronisation, not signal strength. Redundancy changes the picture — at $R = 16, \kappa = 0.05$ and 1024 tokens ($n = 16$) a random 5% edit leaves TPR at 0.88 and a random 10% edit halves it to 0.50, with false-positive control holding on identically attacked controls — and so does locality: because damage propagates forward only, re-denoising one contiguous span of 10% or even 30% of positions leaves the pooled detector exactly at its clean 0.88; what an edit costs is set by where it starts, not by how much it changes. We also measured the two repairs the pooled count seems to invite. A per-block exact test combined by Bonferroni keeps the document-level guarantee by union bound yet never beats pooling (0.38 against 0.50 under the random 10% attack, 0.69 against 0.88 clean), because uniformly random edits damage every block and the sparse-signal regime where concentration-type tests win [27] never arises; a 128-token conditioning window, which bounds an edit's reach to five blocks, costs nothing clean (0.69 in both arms at 512 tokens, $n = 16$) but buys nothing under random 10% re-denoising (0.44 against 0.50) for the same reason. The one untested regime that could still favour a min-statistic is copy–paste-style sparse survival.

5 Conclusion

Shibboleth argues provenance from a keyed probe that a masked DLM answers about its own output rather than from a pattern left in the tokens, and because the gate is settled before the key names the sign it accepts, carriers may be discovered adaptively at no cost to the detector's exact p-values. The trade pays where the decoding schedule must not change, where the evidence wanted is one a specified model can vouch for, which a key–token hash cannot supply, and, as the two-model comparison shows, where a token-local channel's strength would otherwise hinge on the decoder behaviour it exploits: both baselines weakened sharply when carried unchanged from LLaDA to Dream while the probe travelled with the model. Where a token-local watermark does bite, it stays cheaper and easier to resynchronise after edits.

Limitations. That guarantee is one-sided: soundness holds for any text written without the key, nothing is claimed in the other direction, and dispersed editing degrades detection while contiguous edits cost nothing (Section 4.2), so Shibboleth claims resilience to light or localized editing at high redundancy and no more. Verification costs 144 masked evaluations per document, ruling out corpus screening, and is bound to one checkpoint and precision. The evaluation is narrow (two masked DLMs, one prompt corpus, cohorts of 16–50 documents, provenance only; the payload construction of Section 3.5 is unmeasured); Appendix C lists repairs.

References

[1] Scott Aaronson and Hendrik Kirchner. Watermarking GPT outputs, 2023. https://www.scottaaronson.com/talks/watermark.ppt.

[2] Marianne Arriola, Subham Sekhar Sahoo, Aaron Gokaslan, Zhihan Yang, Zhixuan Qi, Jiaqi Han, Justin T. Chiu, and Volodymyr Kuleshov. Block diffusion: Interpolating between autoregressive and diffusion language models. In *ICLR*, 2025.

[3] Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured denoising diffusion models in discrete state-spaces. In *NeurIPS*, 2021.

[4] Arka Bagchi, Akhil Bhimaraju, Moulik Choraria, Daniel Alabi, and Lav R. Varshney. Watermarking discrete diffusion language models. *arXiv preprint arXiv:2511.02083*, 2025.

[5] Mihir Bellare, Ran Canetti, and Hugo Krawczyk. Keying hash functions for message authentication. In *CRYPTO*, 1996.

[6] Heli Ben-Hamu, Itai Gat, Daniel Severo, Niklas Nolte, and Brian Karrer. Accelerated sampling from masked diffusion models via entropy bounded unmasking. In *NeurIPS*, 2025.

[7] Adam Block, Alexander Rakhlin, and Ayush Sekhari. GaussMark: A practical approach for structural watermarking of language models. In *ICML*, 2025.

[8] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.

[9] Ruibo Chen, Yihan Wu, Yanshuo Chen, Chenxi Liu, Junfeng Guo, and Heng Huang. A watermark for order-agnostic language models. In *ICLR*, 2025.

[10] Miranda Christ, Sam Gunn, and Or Zamir. Undetectable watermarks for language models. In *COLT*, 2024.

[11] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

[12] Pierre Fernandez, Antoine Chaffin, Karim Tit, Vivien Chappelier, and Teddy Furon. Three bricks to consolidate watermarks for large language models. In *2023 IEEE International Workshop on Information Forensics and Security (WIFS)*, 2023.

[13] Jiayi Fu, Xuandong Zhao, Ruihan Yang, Yuansen Zhang, Jiangjie Chen, and Yanghua Xiao. GumbelSoft: Diversified language model watermarking via the GumbelMax-Trick. In *ACL*, 2024.

[14] Thibaud Gloaguen, Robin Staab, Nikola Jovanović, and Martin Vechev. Watermarking diffusion language models. In *ICLR*, 2026. arXiv:2509.24368.

[15] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *ICLR*, 2021.

[16] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *NeurIPS*, 2020.

[17] Pyo Min Hong and Albert No. dgMARK: Decoding-Guided Watermarking for Diffusion Language Models. In *ICML*, 2026. arXiv:2601.22985.

[18] Zhengmian Hu, Lichang Chen, Xidong Wu, Yihan Wu, Hongyang Zhang, and Heng Huang. Unbiased watermark for large language models. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.

[19] Nikola Jovanović, Robin Staab, and Martin Vechev. Watermark stealing in large language models. In *ICML*, 2024.

[20] Jaeyeon Kim, Kulin Shah, Vasilis Kontonis, Sham M. Kakade, and Sitan Chen. Train for the worst, plan for the best: Understanding token ordering in masked diffusions. In *ICML*, 2025.

[21] John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. A watermark for large language models. In *ICML*, 2023.

[22] John Kirchenbauer, Jonas Geiping, Yuxin Wen, Manli Shu, Khalid Saifullah, Kezhi Kong, Kasun Fernando, Aniruddha Saha, Micah Goldblum, and Tom Goldstein. On the reliability of watermarks for large language models. In *ICLR*, 2024.

[23] Kalpesh Krishna, Yixiao Song, Marzena Karpinska, John F. Wieting, and Mohit Iyyer. Paraphrasing evades detectors of AI-generated text, but retrieval is an effective defense. In *NeurIPS*, 2023.

[24] Rohith Kuditipudi, John Thickstun, Tatsunori Hashimoto, and Percy Liang. Robust distortion-free watermarks for language models. *TMLR*, 2024.

[25] Inception Labs, Samar Khanna, Siddhant Kharbanda, Shufan Li, Harshit Varma, Eric Wang, Sarah Birnbaum, Ziyang Luo, Yanis Miraoui, and Akash Palrecha. Mercury: Ultra-fast language models based on diffusion. *arXiv preprint arXiv:2506.17298*, 2025.

[26] Chia-Ming Lee, Shao-Kai Liu, Ming-Ching Chang, Xin Li, Yu-Lun Liu, and Chih-Chung Hsu. Commit locally, exit globally: Coordinating adaptive sampling and early exit in diffusion language models, 2026. URL https://arxiv.org/abs/2607.28166.

[27] Xiang Li, Feng Ruan, Huiyuan Wang, Qi Long, and Weijie J. Su. Robust detection of watermarks for large language models under human edits. *Journal of the Royal Statistical Society: Series B*, 2025.

[28] Yuqing Liang, Jiancheng Xiao, Wensheng Gan, and Philip S. Yu. Watermarking techniques for large language models: A survey. *Artificial Intelligence Review*, 2026.

[29] Aaron Lou, Chenlin Meng, and Stefano Ermon. Discrete diffusion modeling by estimating the ratios of the data distribution. In *ICML*, 2024.

[30] Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. Large language diffusion models. In *NeurIPS*, 2025.

[31] Jingyang Ou, Shen Nie, Kaiwen Xue, Fengqi Zhu, Jiacheng Sun, Zhenguo Li, and Chongxuan Li. Your absorbing discrete diffusion secretly models the conditional distributions of clean data. In *ICLR*, 2025.

[32] Fabien A. P. Petitcolas, Ross J. Anderson, and Markus G. Kuhn. Information hiding — a survey. In *Proceedings of the IEEE*, 1999.

[33] Ohad Raban, Ethan Fetaya, and Gal Chechik. LR-DWM: Efficient watermarking for diffusion language models. *arXiv preprint arXiv:2601.12376*, 2026.

[34] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. Technical report, OpenAI, 2019.

[35] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

[36] Subham Sekhar Sahoo, Marianne Arriola, Aaron Gokaslan, Edgar Marroquin Marroquin, Alexander M. Rush, Yair Schiff, Justin T. Chiu, and Volodymyr Kuleshov. Simple and effective masked diffusion language models. In *NeurIPS*, 2024.

[37] Jiaxin Shi, Kehang Han, Zhe Wang, Arnaud Doucet, and Michalis Titsias. Simplified and generalized masked diffusion for discrete data. In *NeurIPS*, 2024.

[38] Qingyan Wei, Yaojie Zhang, Zhiyuan Liu, Pengfei Zeng, Yu Wang, Biao Qi, Dong Liu, and Linfeng Zhang. Accelerating diffusion large language models with SlowFast sampling: The three golden principles. In *ICLR*, 2026.

[39] Chengyue Wu, Hao Zhang, Shuchen Xue, Zhijian Liu, Shizhe Diao, Ligeng Zhu, Ping Luo, Song Han, and Enze Xie. Fast-dLLM: Training-free acceleration of diffusion LLM by enabling KV cache and parallel decoding. In *ICLR*, 2026.

[40] Linyu Wu, Linhao Zhong, Wenjie Qu, Yifan Li, Yue Liu, Shengfang Zhai, Chunhua Shen, and Jiaheng Zhang. DMark: Order-agnostic watermarking for diffusion large language models. *arXiv preprint arXiv:2510.02902*, 2025.

[41] Jiacheng Ye, Zhihui Xie, Lin Zheng, Jiahui Gao, Zirui Wu, Xin Jiang, Zhenguo Li, and Lingpeng Kong. Dream 7B: Diffusion large language models. *arXiv preprint arXiv:2508.15487*, 2025.

[42] Xuandong Zhao, Yu-Xiang Wang, and Lei Li. Protecting language generation models via invisible watermarking. In *ICML*, 2023.

[43] Xuandong Zhao, Prabhanjan V. Ananth, Lei Li, and Yu-Xiang Wang. Provable robust watermarking for AI-generated text. In *ICLR*, 2024.

[44] Fengqi Zhu, Rongzhen Wang, Shen Nie, Xiaolu Zhang, Chunwei Wu, Jun Hu, Jun Zhou, Jianfei Chen, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. LLaDA 1.5: Variance-reduced preference optimization for large language diffusion models. *arXiv preprint arXiv:2505.19223*, 2025.

Nominal α	Mean FPR	Worst-key FPR	Verdict
0.10	0.0121	0.0583	valid ($\leq \alpha$)
0.05	0.0037	0.0250	valid ($\leq \alpha$)
0.01	0.0000	0.0000	valid ($\leq \alpha$)

Table 3: **False positive control holds per key, worst key included.** 20 keys $\times$ 120 non-watermarked generations under the deployed rule.

		TPR at FPR ↑	
Attack	Strength	1%	0.1%
Weakest setting: $R = 1$, *512 tokens* ($n = 20$)			
None	—	0.35	??
Re-denoising (same model, argmax)	10% of positions	0.05	??
Deletion	10% of positions	0.05	??
Insertion	10% of positions	??	??
Substitution (random)	10% of positions	??	??
Copy–paste	25% kept, 1 segment	??	??
Paraphrase (DIPPER) [23]	lex 20, order 0	??	??
Strongest setting: $R = 16, \kappa = 0.05$, *1024 tokens* ($n = 16$)			
None	—	0.88	0.88
Re-denoising (random positions)	5% of positions	0.88	0.75
	10% of positions	0.50	0.38
Re-denoising (contiguous span)	10% of positions	0.88	0.88
	30% of positions	0.88	0.88

Table 4: **Robustness to post-editing.** Detection after each attack under the deployed pooled detector, scored against the same analytic thresholds as Table 1. Evidence written into a replayable probe is synchronised to the prefix, so damage propagates forward only: at $R = 1$ a 10% edit invalidates nearly every carrier, while at the strongest setting redundancy absorbs a random 5% edit outright, a random 10% edit halves detection, and a contiguous edit costs nothing even at 30% of positions, because every carrier before the span survives untouched. Small cohorts: one document is 0.05 ($n = 20$) or 0.06 ($n = 16$) of TPR.

A Additional Tables

Table 5 is the full quality grid, including the greedy-decoding cells that the compact Table 2 omits. Table 3 reports the false-positive rate of the deployed decision rule measured per key, the empirical counterpart of Proposition 1 discussed in Section 4.2. Table 4 lists every post-editing attack measured, at the weakest and the strongest setting, under the deployed pooled detector; Section 4.2 discusses the block-local and windowed alternatives that were measured against it.

		Greedy decoding			Multinomial decoding		
Model	Method	MMLU Acc ↑	GSM8K Acc ↑	HumanEval Pass@1 ↑	MMLU Acc ↑	GSM8K Acc ↑	HumanEval Pass@1 ↑
LLaDA-8B-Instruct	Non-watermarked	0.605	0.740	0.409	0.595	0.640	0.335
	KGW [21]	??	??	??	0.535	0.740	0.293
	dgMARK [17]	??	??	??	0.530	0.660	??
	dgMARK +3-beam	??	??	??	0.530	0.660	??
	Shibboleth	??	??	??	0.575	0.700	0.384
Dream-7B-Instruct	Non-watermarked	0.740	0.840	??	0.695	0.660	??
	KGW	??	??	??	0.625	0.700	??
	dgMARK	??	??	??	0.570	0.560	??
	dgMARK +3-beam	??	??	??	0.645	0.560	??
	Shibboleth	??	??	??	0.710	0.680	??

Table 5: **Text generation quality, full grid.** Accuracy on MMLU [15] and GSM8K [11] and pass@1 on HumanEval [8] under greedy and multinomial decoding, with the unwatermarked run of each model as the reference row of its block. Shibboleth uses $R = 16, \kappa = 0.05$; 50 problems per GSM8K cell, so differences within ± 0.1 are noise. Each watermark decodes under its own regime — KGW left-to-right (its green list requires a resolved prefix; its own control scores 0.640), dgMARK under confidence decoding (its argmax control is the greedy reference cell) — so rows are read against their own controls, not across. Cells marked **??** are pending measured runs.

B Post-hoc Substitution

An embedder that swapped tokens in the finished text would be cheaper than one intervening during sampling, and we begin by measuring how much room such a swap has. Write τ for the likelihood budget of a substitution, the number of nats by which a replacement may fall below the live argmax. Across 36 settings that replace tokens after ordinary generation, the median selected position admits exactly one token at $\tau = 3$, the mean rising only from 1.6 at $\tau = 3$ to 5.6 at $\tau = 6$, and the best setting inside a $1.35\times$ perplexity budget reaches no more than 0.10 TPR at 1% FPR. Conditionals this sharp leave too little room for a forced replacement, and the embedder must therefore write while the conditional is still live.

C Future Work

Each boundary suggests its own repair. Anchoring probe patterns to content rather than to absolute positions would stop an edit from desynchronising every carrier after it, which is the single change most likely to turn an abrupt collapse into a graceful decline; the attack still missing is copy–paste-style sparse survival, the one measured regime left where a per-block min-statistic could yet beat the pooled count. Verification cost is the easier target, since the binomial tail is monotone in the matches so far and a sequential test could stop once the evidence is decisive. Loosening the checkpoint requirement in turn asks for a contrast quantised coarsely enough that numerical drift cannot flip a sign, trading a little capacity for portability. Adaptive adversaries deserve their own study, since a detector answering queries leaks something about its key. Table 1 finally argues for composition rather than a winner, with a cheap token-local screen deciding what is worth a model-response check.